# AQ1.4: Activity Questions 4 - Not Graded

This assignment will not be graded and is only for practice.

**4.1 Level 1:**

Let $A$ be a $3 \times 3$ matrix with non-zero determinant. If $det(2A) = k\ det(A)$, then what will be the value of $k$?

[Hint: If a scalar $(c)$ is multiplied with one row of a matrix $A$, then the determinant of the new matrix will be $c$ times the determinant of $A$.]

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 8

*1 point*

*1 point*

If $A = \begin{bmatrix} 2 & -1 & 1 \\ 1 & -3 & 4 \\ -1 & 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 3 & -2 \\ 0 & 1 & -1 \\ 3 & 4 & 2 \end{bmatrix}$, then choose the set of correct options.

- [ ] $det(A) = -9$ and $det(B) = -3$.
- [ ] $det(A) = 9$ and $det(B) = 3$.
- [ ] $det(A) = -9$ and $det(B) = 3$.
- [ ] $det(AB) = -27$ and $det(BA) = 27$.
- [ ] $det(AB) = det(BA) = -27$.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$det(A) = -9$ and $det(B) = 3$.
$det(AB) = det(BA) = -27$.

*1 point*

Let $A$ be a $2 \times 2$ matrix, which is given as $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$. Define the following matrices :

$$B = \begin{bmatrix} a_{11} - a_{21} & a_{12} - a_{22} \\ a_{21} & a_{22} \end{bmatrix}, \quad C = \begin{bmatrix} a_{11} - a_{12} & a_{12} \\ a_{21} - a_{22} & a_{22} \end{bmatrix},$$

$$D = \begin{bmatrix} a_{11} + a_{21} & a_{12} - a_{22} \\ a_{21} & a_{22} \end{bmatrix}, \quad E = \begin{bmatrix} a_{11} - a_{21} & a_{12} + a_{22} \\ a_{21} & a_{22} \end{bmatrix}$$

Which of the matrices among $B, C, D,$ and $E$ have the same determinant as that of the matrix $A$, for any real numbers $a_{11}, a_{12}, a_{21}, a_{22}$?

○ $B$ and $D$

○ $B$ and $E$

○ $B$ and $C$

○ $D$ and $E$

○ $C$ and $E$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$B$ and $C$

Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ ta_{11} - sa_{31} & ta_{12} - sa_{32} & ta_{13} - sa_{33} \\ ra_{31} & ra_{32} & ra_{33} \end{bmatrix}$ be a matrix and $r, s, t \neq 0$. Find $det(A)$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 0
*1 point*

Suppose $A = \begin{bmatrix} -1 & 4 & -2 \\ 0 & 1 & 3 \\ 1 & 0 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 & -1 \\ 4 & \alpha & 0 \\ 0 & -3 & -5 \end{bmatrix}$. If $det(AB) = 32$, then find the value of $\alpha$.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 1
*1 point*

**4.2 Level 2:**

**(Use this information to answer questions 6 and 7)**

Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be the vertices of a triangle. The area of the triangle $ABC$ is given by $\frac{1}{2}|det(D)|$ square units, where

$D = \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}$

If $P(-1, 1)$, $Q(1, 1)$ and $R(1, 0)$ are the vertices of a triangle, then find the area (in square units) of the triangle $PQR$.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 1

*1 point*

*1 point*

Let $P(x-1, 2x)$, $Q(0, x-2)$ and $R(-x+1, 1)$ be the vertices of a triangle. If the area of the triangle is 5 square units and $x > 0$, then choose the set of correct options

- [ ] length of side $PQ = \sqrt{29}$ units.
- [ ] length of side $PR = \sqrt{40}$ units.
- [ ] length of side $QR = 2$ units.
- [ ] length of side $QR = 1$ units.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
length of side $PQ = \sqrt{29}$ units.
length of side $QR = 2$ units.

*1 point*

Find out the correct value of $det(A)$, where $A = \begin{bmatrix} 1 & 2010 & 2020 \times 2030 \\ 1 & 2020 & 2030 \times 2010 \\ 1 & 2030 & 2010 \times 2020 \end{bmatrix}$

[[ Hint: Observe that the given matrix is of the form: $\begin{bmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{bmatrix}$ ]]

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200200

−2000−2000

20002000

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
20002000

*1 point*

Let $A$A be a square matrix of order 33 and $B$B be a matrix that is obtained by adding 22 times the first row of $A$A to the third row of $A$A and adding 33 times the second row of $A$A to the first row of $A$A. What is the value of $det(6A^2 B^{-1})$det(6A2B−1)?

6 $det(A)$6 det(A)

6 $det(A)det(B)$6 det(A)det(B)

$6^3\ det(A)$63 det(A)

$6^3\ det(A)det(B)$63 det(A)det(B)

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$6^3\ det(A)$63 det(A)

Suppose $A = \begin{bmatrix} 19 & 20 & 21 \\ 20 & 21 & 19 \\ 21 & 19 & 20 \end{bmatrix}$A = 192021202119211920. Find the value of $\frac{1}{60} det(A^2)$601det(A2).

[[ You can try to compute the determinant of a matrix of the form: $\begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$abcbcacab ]]

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 540

*1 point*